

Phase V2.2 Backend Quality Report

Executive Summary

- Version under review: `2.2`
- Objective: finalize backend stabilization, persistence, quality-gate packaging, and readiness assessment
- Automated local regression suite: `18 passed, 0 failed` on August 8, 2026.
- Local health check: `/health` returned `healthy` with version `2.2`.
- Background metadata persistence is now available with local JSON fallback and optional Supabase RE

Architecture

- Endpoint inventory entries: `58`
- Engine inventory components: `16`
- Synchronous generation endpoints remain intact.
- Background endpoints remain additive and backward compatible.

Regression Results

- Local regression suite covered admin analytics, background jobs, health checks, safe errors, idempoten
- Existing synthetic evidence from V2.0 and V2.1 was reused for quality and long-running workflow valida

Resume Quality Results

- Resume Quality score: `82`
- ATS Quality score: `50`
- Recruiter Readability score: `76`
- Existing synthetic resume outputs show grouped skills, no placeholder leakage in reviewed manual sam

Document Results

- Document Rendering score: `82`
- Representative PDF and DOCX artifacts were copied into the final manual-review package.

Performance

- Performance score: `76`
- Long-running synchronous V2.0 baseline examples: portfolio ~`14` ms, job application ~`16` ms.
- Background V2.1 total processing examples: portfolio ~`114759` ms, job application ~`181586` ms.
- Background jobs improve user experience and timeout resilience even when total AI processing time re

Security

- Security score: `88`
- Privacy score: `90`
- API Reliability score: `100`
- Backward Compatibility score: `93`
- Admin analytics remain protected with secret-gated access.
- Health endpoints and background-job metrics are available without exposing secrets or full resume cont

Background Jobs

- Background Job Reliability score: `84`
- Metadata now survives restart where persistence is available.
- Historical background-job analytics are now persisted separately from volatile in-memory summaries.

Manual Review Package

- `graduate\_vlsi`: 2 file(s)
- `graduate\_devops`: 2 file(s)
- `business\_analyst`: 2 file(s)
- `career\_switcher`: 2 file(s)
- `restaurant\_manager`: 2 file(s)
- `senior\_software\_manager`: 2 file(s)

- `cloud\_resume\_optimizer`: 2 file(s)
- `cover\_letter`: 2 file(s)
- `linkedin\_report`: 2 file(s)
- `interview\_report`: 2 file(s)
- `portfolio\_sample`: 4 file(s)
- `job\_application\_report`: 2 file(s)

Known Issues

- Background job restart recovery returns artifact-backed results where available, but job-application restarts are not resumable mid-phase
- Background execution still uses in-process workers in this phase, so active jobs are not resumable mid-phase
- Some historical resume benchmark scores remain moderate for ATS alignment, especially legacy synth

Final Backend Readiness Decision

BACKEND READY WITH MINOR ISSUES

The backend is stable enough to freeze for frontend redesign, but the restart-recovery gap for full job-app